CSET485 – AI and Society: Assignment #1

Stress-Testing a Recidivism Risk Model (COMPAS)

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Dataset: COMPAS Recidivism Risk Scores | random_state = 172 | 70/30 train-test split

Notebook link: https://github.com/RajyavardhanSinghRathore/AI-and-Society-Assignment

0. Data Preprocessing

Data and filters. The ProPublica document compas-scores-two-years.csv consists of approximately 7,214 rows and 53 columns. Then, I applied the usual ProPublica filters to the data file, step by step. The first filter removed only 1,042 rows from the main data file. The three remaining filters did not work at all. Of the 6,172 rows that remained, 45.5% reoffended.

Filter (applied in order)	Rows removed	Rows remaining
days_b_screening_arrest missing or outside [-30, 30]	1,042	6,172
is_recid == -1 (outcome not recorded)	0	6,172
c_charge_degree == 'O' (ordinary traffic offence)	0	6,172
score_text == 'N/A'	0	6,172

Missing values. After filtering there are no missing values in the seven features, the two targets or race, so I did not impute anything. Rows with a missing days_b_screening_arrest were dropped by the filter instead of being filled in.

Targets. Model A predicts decile_score (1–10). Model B predicts two_year_recid (1 = re-offended within two years), which I call is_recidivist. I did not use is_recid because it has no two-year limit. decile_score is never used as an input.

Encoding and features. sex became sex_male (Male = 1, Female = 0) and charge degree became charge_felony (Felony = 1, Misdemeanour = 0). The seven features are age, sex_male, priors_count, juv_fel_count, juv_misd_count, juv_other_count and charge_felony. I dropped age_cat because it is just age put into bins. Race is not a model input (as in the real COMPAS tool). I only use it to make subgroups in Parts
2.1 and 2.4, and once as dummy variables (Caucasian = reference) in a side check in Part 2.2. Race counts: African-American 3,175, Caucasian 2,103, Hispanic 509, Other 343, Asian 31, Native American 11.

Scaling. Both models are unpenalised (OLS and maximum-likelihood logistic regression), so scaling the inputs does not change the predictions. I checked this: a logistic model on standardised features gives the same test probabilities as the raw one (largest difference 2.2e-16), and scikit-learn's unpenalised
LogisticRegression matches my Newton–Raphson fit to within 9.1e-05. So I kept the natural units (years and counts), which keeps the coefficients easy to read.

Outliers. I used Tukey's 1.5×IQR rule. The outliers are real values, not typing mistakes (for example one person has 38 priors), and they are the high-risk cases the model should handle, so I kept them.

Feature	Min	Max	# outliers (IQR)	% of rows
age	18	96	51	0.8
priors_count	0	38	492	8.0
juv_fel_count	0	20	208	3.4
juv_misd_count	0	13	352	5.7
juv_other_count	0	9	461	7.5

As a check, I capped priors_count at its 99th percentile (22). Test accuracy of the logistic model went from 67.87% to 67.76%, so the outliers are not driving the results.

Split. 70/30 split with random_state = 172 and no stratification: 4,320 training rows and 1,852 test rows. The test re-offending rate is 43.7%. Part 1, 2.3 and 2.4 are trained on the training set and checked on the test set. The two inference parts (2.1 and 2.2) use all 6,172 cleaned rows to get steadier estimates. Standard errors, 95% confidence intervals and p-values are Wald statistics from the logistic Hessian.

Part 1 – Build

Both models use the same seven features. Model A is OLS on decile_score (test $R^2 = 0.403$, RMSE = 2.19 decile points). Model B is logistic regression on two_year_recid (67.9% test accuracy at $p \geq 0.5$). Coefficients from the training set:

Feature	OLS coef	OLS SE	Logit coef	Logit SE	Logit p	Odds ratio

intercept	6.815	0.134	0.393	0.138	0.004	1.48
age	-0.106	0.003	-0.044	0.003	<0.001	0.96
sex_male	-0.065	0.084	0.366	0.087	<0.001	1.44
priors_count	0.281	0.007	0.162	0.010	<0.001	1.18
juv_fel_count	0.426	0.084	0.069	0.107	0.520	1.07
juv_misd_count	0.193	0.076	-0.114	0.088	0.195	0.89
juv_other_count	0.328	0.068	0.299	0.081	<0.001	1.35
charge_felony	0.477	0.069	0.190	0.070	0.007	1.21

Thresholds. Model A's test predictions are cut at their median (4.491): at or above the median is "high risk". Model B's probabilities are cut at 0.5.

	B: High risk (p ≥ 0.5)	B: Low risk (p < 0.5)	Total
A: High risk (≥ median)	678	248	926
A: Low risk (< median)	14	912	926
Total	692	1160	1852

Mismatch: 262 of 1,852 people (14.15%) were flagged differently. 248 (13.39%) were high-risk under A but low-risk under B, and 14 (0.76%) the other way round. The two methods agreed on 1,590 people (85.85%).

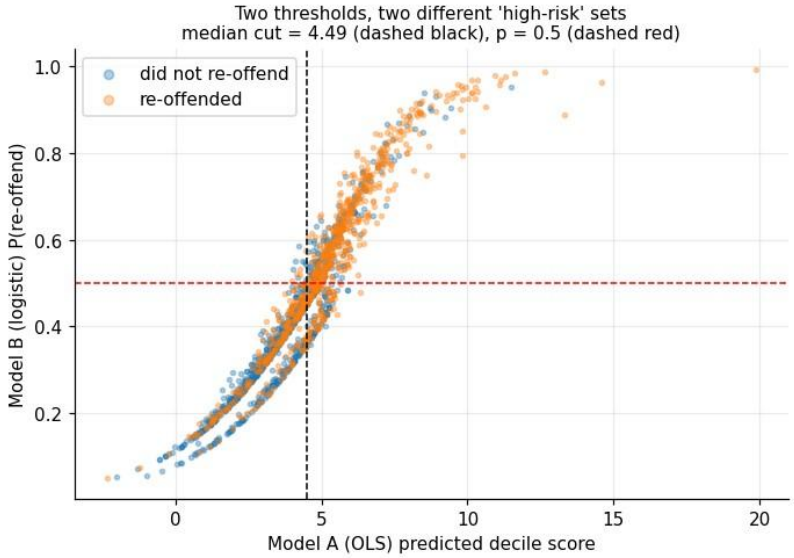


Figure 1. Each dot is one test person. The dashed lines are the two cut-offs. Mismatches are the points in the upper-left and lower-right areas, almost all in the lower-right (A high, B low).

Why do the two approaches differ? (1) The median rule always flags 50% of people, while the 0.5 rule flags 37.4%. (2) The two scores rank people almost the same way (Spearman 0.956), so most of the difference comes from the cut-offs. (3) A linear model on a bounded score goes outside the range (predictions from -2.36 to 19.86), while the logistic curve does not. (4) They predict different things: COMPAS's own score versus actual re-offending. Details are in Part 3, Q1.

Part 2 – The Trap

2.1 Simpson's Paradox Hunt

Split: race. Subgroup A = African-American (n = 3,175) and subgroup B = all other defendants (n = 2,997), so A and B together are the full data. I ran logistic regression on the full data, on A and on B (log-odds coefficients with 95% CI). The last columns test whether the A and B coefficients differ (z-test on the difference).

Predictor	Full data	A: African-American	B: all others	A – B	z	p (diff)
juv_other_count	0.274 [0.135, 0.413]	0.553 [0.329, 0.778]	0.053 [-0.121, 0.226]	0.500	3.45	<0.001

juv_misd_count	-0.012 [-0.167, 0.143]	-0.101 [-0.279, 0.077]	0.126 [-0.180, 0.432]	-0.227	-1.26	0.208
priors_count	0.163 [0.147, 0.178]	0.150 [0.130, 0.170]	0.190 [0.161, 0.219]	-0.040	-2.24	0.025
age	-0.044 [-0.049, -0.038]	-0.050 [-0.059, -0.042]	-0.036 [-0.044, -0.029]	-0.014	-2.47	0.014
sex_male	0.326 [0.185, 0.467]	0.522 [0.320, 0.724]	0.124 [-0.074, 0.321]	0.399	2.77	0.006
juv_fel_count	0.072 [-0.105, 0.249]	0.064 [-0.140, 0.268]	0.041 [-0.322, 0.403]	0.023	0.11	0.913
charge_felony	0.228 [0.113, 0.344]	0.187 [0.022, 0.351]	0.246 [0.082, 0.409]	-0.059	-0.50	0.620

Result. No significant coefficient changes its sign, so this is **not a textbook Simpson's reversal**. Instead I report the subgroup where the effect is clearly stronger or weaker. For juv_other_count the log-odds are 0.274 (full), **0.553 in A** and **0.053 in B** (odds ratios 1.32, 1.74 and 1.05). The A–B difference is 0.500 (z = 3.45, p = 0.0006). A likelihood-ratio test of a juv_other_count × race interaction gives LR = 14.99, p = 0.0001 (interaction coefficient 0.535, 95% CI [0.258, 0.812]). I screened 26 (split, predictor) pairs, so the Bonferroni-adjusted p-value is 0.014, which is still below 0.05. For juv_misd_count the sign does change (-0.012 full, -0.101 in A, +0.126 in B), but every CI includes 0 and the difference is not significant (p = 0.208), so I do not call it a reversal. Other significant differences are sex_male (0.522 vs 0.124, p = 0.006), age (p = 0.014) and priors_count (p = 0.025).

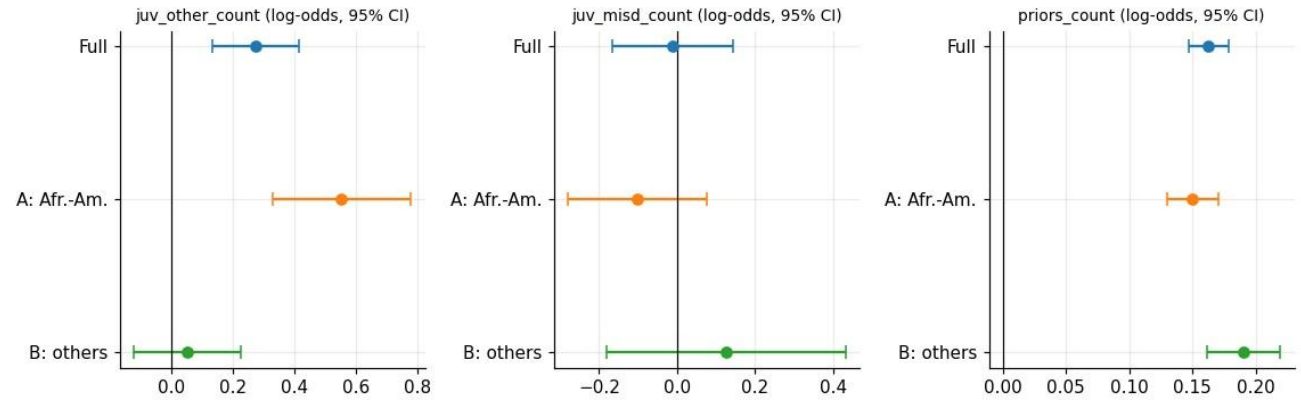


Figure 2. Coefficients and 95% CIs for the full data and the two subgroups. For juv_other_count the interval for group B crosses zero, while group A's is far to the right.

Why does the effect differ? The pooled coefficient (0.274) is a blend of two different relationships. 9.2% of group A (293 people) but only 5.6% of group B (168 people) have a juvenile "other" record, so A dominates the pooled number. In the raw data, re-offending goes from 49.5% to 80.2% in A when there is such a record, and from 37.2% to 57.1% in B. After adjusting for age and priors, B's coefficient (0.053, p = 0.552) is not distinguishable from zero. One possible reason (I did not test it) is that minor juvenile contacts are recorded more at the officer's discretion, so the record says more about policing than behaviour in one group.

Policy implication. If the subgroups are ignored, the conclusion would be "a juvenile 'other' record raises odds by 32% for everyone". In reality it raises odds by about 74% in A and by an undetectable 5% in B. A single weight would over-penalise group B and could build policing patterns into everyone's score. The right conclusion is "audit or drop this feature", not "use it for all".

2.2 Omitted-Variable Bias

Dropped variable: priors_count. It is positively correlated with the other predictors (r = 0.27 with juv_misd_count, 0.19 with juv_fel_count, 0.15 with charge_felony, 0.12 with age, 0.12 with sex_male) and it is the strongest single driver of re-offending (full-data coefficient 0.163), so leaving it out should push its signal onto the others. Percentage change = (after − before) / |before|.

Predictor	Before (full)	After (reduced)	Abs. change	% change	Shift in SEs	p before → after
age	-0.044	-0.028	+0.015	+34.9%	+5.58	<0.001 → <0.001
sex_male	0.326	0.451	+0.125	+38.4%	+1.74	<0.001 → <0.001
juv_fel_count	0.072	0.441	+0.369	+512.3%	+4.07	0.427 → <0.001
juv_misd_count	-0.012	0.387	+0.399	+3,303.0%	+5.06	0.878 → <0.001

juv_other_count	0.274	0.380	+0.106	+38.6%	+1.49	<0.001 → <0.001
charge_felony	0.228	0.409	+0.181	+79.0%	+3.07	<0.001 → <0.001
intercept	0.354	0.067	-0.287	-81.1%	–	0.002 → 0.543
race_aa (side model with race dummies)	0.063	0.313	+0.249	+394.5%	+3.94	0.318 → <0.001

Which coefficient shifted the most? By percentage it is juv_misd_count (from -0.012 to 0.387, +3,303%), but this is inflated because it started almost at zero. In standard errors, age moved the most (+5.58 SEs), then juv_misd_count (+5.06) and juv_fel_count (+4.07, +512%). Two predictors went from "not significant" to "highly significant" (juv_fel_count p 0.427 → <0.001 and juv_misd_count p 0.878 → <0.001). In the side model the African-American dummy rises from 0.063 (p = 0.318) to 0.313 (p <0.001). Test accuracy also falls from 67.9% to 62.4%.

Lesson: a coefficient only means something given the other variables in the model. Without this test one would wrongly say that juvenile misdemeanours are a strong independent predictor (0.387), when the adjusted effect is almost zero (-0.012). A real-world audit for hidden confounds is described in Part 3, Q4.

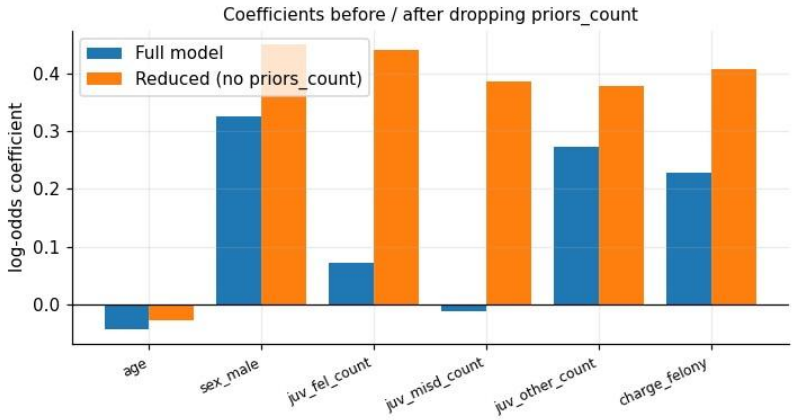


Figure 3. Coefficients with and without priors_count.

2.3 Adversarial Subset Construction

On the test set, Model B gives P ≥ 0.75 to 149 people and P ≤ 0.25 to 258 people. The confident **and** wrong cases are 28 false positives (P ≥ 0.75, actual 0) plus 55 false negatives (P ≤ 0.25, actual 1) = **83 cases, 4.48% of the 1,852 test rows** (18.8% of the confident positives are wrong, and 21.3% of the confident negatives). Since 83 is more than 50, I took a random sample of 50 (random_state = 172): 21 false positives and 29 false negatives.

Characteristic	Adversarial 50	Full test set (n = 1,852)
Mean age (years)	46.9	34.5
% aged ≥ 30	92.0	56.4
% aged < 25	2.0	22.0
% aged > 45	52.0	18.9
% male	80.0	80.5
% African-American	56.0	49.6
% Caucasian	34.0	35.4
Mean priors_count	6.70	3.15
% with zero priors	32.0	34.1
% with ≥ 5 priors	42.0	23.4
% felony charge	62.0	64.6
% with any juvenile record	20.0	12.8
Mean predicted probability	0.462	0.458

Common characteristic: age. 92% of the 50 rows (46 of 50) are aged 30 or older, against 56.4% of the test set. 52% are over 45 (vs 18.9%) and only 2% are under 25 (vs 22.0%). Sex and race are close to the test-set mix. There are two kinds of errors inside the sample: the false negatives are older (mean 55.0 years, 55% with zero priors), and the false positives have very long records (mean 14.8 priors).

Across all 83 confident errors, 76 (91.6%) are aged 30 or more. The confident-error rate is 21.4% for ages 30 and over versus 13.5% under 30.

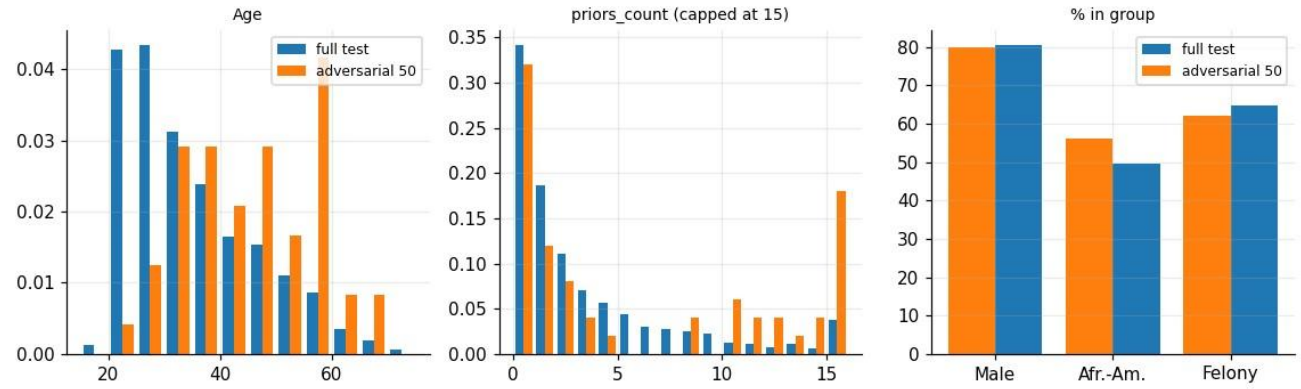


Figure 4. Age, priors and group mix of the 50 adversarial rows versus the full test set.

Calibration of the logistic model by number of priors (training set):

priors_count	n	Observed re-offend rate	Mean predicted P
0	1453	0.288	0.342
1	783	0.405	0.393
2-3	811	0.505	0.451
4-6	544	0.588	0.534
7-10	373	0.713	0.646
11+	356	0.756	0.836

Why does the model fail confidently here? (i) *Noise that cannot be removed:* a calibrated model that says P ≤ 0.25 (mean predicted 0.184) will still be wrong in about 18% of those cases; the observed re-offending rate there is 21.3%. (ii) *Over-confidence at the top:* for P ≥ 0.75 the mean prediction is 0.852 but only 81.2% re-offend. (iii) *Wrong shape:* the model is linear on the log-odds scale in both age and priors. The true effect of priors is curved: adding log(1+priors) improves the fit (LR = 64.1, p = 1.2e-15), and the table above shows over-prediction at 0 priors (0.342 vs 0.288) and at 11+ priors (0.836 vs 0.756) but under-prediction at 2–10. A missing age × priors interaction is also significant (coefficient -0.0022, LR = 9.38, p = 0.0022). The negative sign means each extra prior raises risk less for older people than the model assumes, so it overstates risk for older people with long records (false positives), while its fixed age penalty (-0.0439 per year) understates it for older people with short records (false negatives).

2.4 Fairness Trade-Off Analysis

Sensitive variable: race. Group A = Caucasian (n = 656 in the test set), Group B = African-American (n = 919). Other races are left out here. Observed test re-offending rates are 37.3% (A) and 49.8% (B); the mean predicted probability is 0.400 vs 0.513. All ratios are A / B, so a value below 1 means group B is flagged (or wrongly flagged) more often. DP = P(Ŷ=1|A)/P(Ŷ=1|B); EO = min(TPR /TPR , FPR /FPR); PP = PPV /PPV .

					A B	A B	A B
t	**Flagged A / B (n)**	**DP**	**EO**	**PP**	**TPR ratio**	**FPR ratio**	**Overall accuracy**
0.05	656 / 919	1.000	1.000	0.749	1.000	1.000	43.7%
0.15	616 / 912	0.946	0.932	0.771	0.974	0.932	45.7%
0.25	512 / 856	0.838	0.804	0.815	0.911	0.804	51.7%
0.35	395 / 742	0.746	0.730	0.830	0.826	0.730	60.4%
0.45	244 / 556	0.615	0.651	0.819	0.672	0.651	66.6%
0.55	109 / 329	0.464	0.507	0.819	0.507	0.575	67.7%
0.65	51 / 208	0.343	0.293	0.982	0.450	0.293	64.6%
0.75	23 / 115	0.280	0.160	1.064	0.398	0.160	61.3%
0.85	13 / 52	0.350	0.374	0.930	0.435	0.374	58.8%
0.95	2 / 11	0.255	0.000	1.100	0.374	0.000	57.0%

Fairness ratios per threshold (ideal = 1.0). At t ≥ 0.85 very few people are flagged (for example 2 Caucasian and 11 African-American defendants at t = 0.95), so those ratios are noisy. Accuracy is on the full 1,852-row test set.

t	Selection A	Selection B	TPR A	TPR B	FPR A	FPR B	PPV A	PPV B
0.05	1.000	1.000	1.000	1.000	1.000	1.000	0.373	0.498
0.15	0.939	0.992	0.971	0.998	0.920	0.987	0.386	0.501
0.25	0.780	0.931	0.882	0.967	0.720	0.896	0.422	0.518
0.35	0.602	0.807	0.759	0.919	0.509	0.696	0.471	0.567
0.45	0.372	0.605	0.522	0.777	0.282	0.434	0.525	0.640
0.55	0.166	0.358	0.273	0.539	0.102	0.178	0.615	0.751
0.65	0.078	0.226	0.159	0.354	0.029	0.100	0.765	0.779
0.75	0.035	0.125	0.082	0.205	0.007	0.046	0.870	0.817
0.85	0.020	0.057	0.041	0.094	0.007	0.020	0.769	0.827
0.95	0.003	0.012	0.008	0.022	0.000	0.002	1.000	0.909

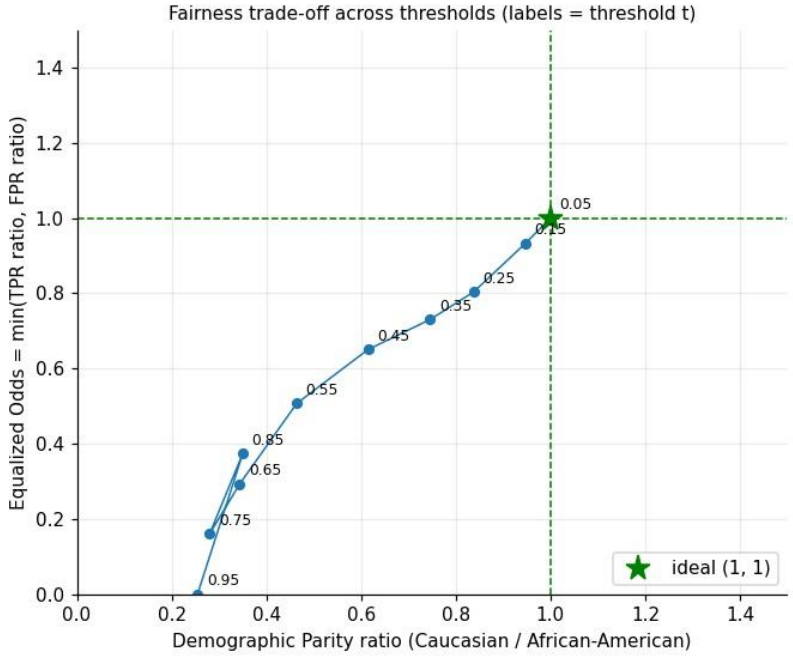


Figure 5. Trade-off chart: x = Demographic Parity, y = Equalized Odds, axes 0–1.5, points labelled with the threshold. The green star is the ideal (1, 1). Up and to the right is fairer. t = 0.05 sits at (1, 1) only because it flags everyone.

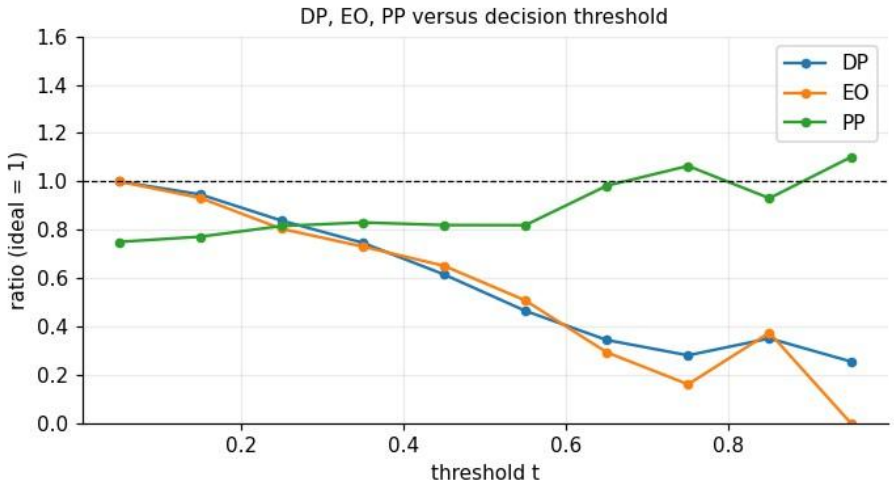


Figure 6. DP, EO and PP as functions of the threshold.

Quantifying the trade-off. In my data DP and EO do not trade off against each other: they rise and fall together as the threshold changes (both are best at the lowest thresholds, where almost everyone is flagged). The real conflict is between DP/EO and (a) predictive parity and (b) usefulness. (1) Improving PP from 0.819 (t = 0.45) to 0.982 (t = 0.65), a +19.8% gain, drops EO from 0.651 to 0.293 (55.0% loss) and DP from 0.615 to 0.343 (44.1% loss). That costs 2.77 (EO) and 2.22 (DP) percentage points lost per point of PP gained. (2) Improving DP from 0.615 to 0.838 (+36.3%, t = 0.45 → 0.25) also improves EO (0.651 → 0.804, +23.6%), but overall accuracy falls from 66.6% to 51.7% (-22.4%). That is below the 56.3% you get by predicting "no re-offence" for everyone, and costs 0.62 accuracy points per DP point gained.

Key insight and recommendation. No threshold satisfies all three. The only thresholds with DP and EO ≥ 0.9 (t ≤ 0.15) flag 94–99% of people and have 45.7% accuracy, while PP is closest to 1 at t ≥ 0.65, where DP and EO have collapsed. I recommend **t = 0.45**: DP 0.615, EO 0.651, PP 0.819, accuracy 66.6% (only 1.1 points below the best accuracy at t = 0.55). It keeps the worst of the three ratios (0.615) higher than every threshold from 0.55 up, and it is still clearly informative. The reasoning is in Part 3, Q7–Q8.

Part 3 – Descriptive Analysis Questions

Question 1. Why did the threshold approach and the logistic approach flag (or not flag) the same people?

Model A (OLS on decile_score, median cut-off) and Model B (logistic on two_year_recid, p ≥ 0.5) disagreed on 262 of the 1,852 test people, a mismatch rate of 14.15%. The disagreement is one-sided: 248 people (13.39%) were high-risk under A but low-risk under B, and only 14 (0.76%) the other way. They agreed on 1,590 people (678 high-risk in both, 912 low-risk in both).

Mostly it comes down to where each rule draws its line. A median cut-off flags exactly half the test set (926 people) whatever the risk in the population, and the median of Model A's predictions is 4.49. Model B's 0.5 rule flags only 692 people (37.4%): the median logistic probability is 0.448, below 0.5, and only 43.7% of the test set actually re-offended. So the median rule flags more people than the 0.5 rule. The scores themselves rank people almost the same (Spearman ρ = 0.956). When I make Model A flag the same share as Model B (cut-off 4.90 instead of 4.49), mismatches fall from 262 to 188 (10.15%). When both use their own medians, there are 171 (9.23%).

The remaining disagreements (about 188) come from model form and target. OLS fits a straight line to a bounded 1–10 score, so its predictions go from -2.36 to 19.86, with 81 test cases (4.37%) outside the valid range, while the logistic curve flattens in the tails. Also, decile_score is COMPAS's own estimate and not the outcome, so Model A copies a score while Model B predicts re-offending. The mismatched people are borderline: their mean P(re-offend) is 0.459 and 43.1% of them re-offended, close to the base rate. Accuracy is 66.4% for A and 67.9% for B.

Question 2. The subgroup with significantly different effects

I split by race: group A (African-American, n = 3,175) and group B (all others, n = 2,997). No key predictor changes sign with statistical support, so I report the subgroup where the effect differs most. For juv_other_count the log-odds coefficient is 0.274 in the full model (95% CI [0.135, 0.413]), 0.553 in A (CI [0.329, 0.778]) and 0.053 in B (CI [-0.121, 0.226]). The odds ratios are 1.32, 1.74 and 1.05. The A–B gap of 0.500 has z = 3.45, p = 0.0006, and the interaction likelihood-ratio test gives p = 0.0001. The sign of juv_misd_count does flip (-0.101 in A, +0.126 in B, -0.012 pooled), but all three CIs include zero (difference p = 0.208), so it is noise and not a Simpson's reversal.

The pooled coefficient is dominated by group A, which has more juvenile "other" records (9.2% vs 5.6%). In the raw data a record goes with higher re-offending in both groups (A: 49.5% → 80.2%; B: 37.2% → 57.1%), but once age and priors are held fixed, B's effect disappears (p = 0.552) while A's stays strong. This fits with minor juvenile contacts being recorded differently across groups, though my data cannot test that.

Policy consequence. If the subgroups were ignored, the pooled estimate would say a juvenile "other" record raises re-offending odds by about 32% for everyone, which would support one uniform weight. The subgroup results say about 74% in A and about 5% (not distinguishable from zero) in B. A uniform weight would over-penalise group B and could build enforcement patterns into the scores for group A. The conclusion changes from "use this feature" to "check it for differences in recording before using it".

Question 3. Variable dropped and omitted-variable shift

I dropped priors_count (number of prior offences). It correlates with every other predictor (r from 0.11 to 0.27) and is the strongest single driver of re-offending (coefficient 0.163, odds ratio 1.18 per prior). The table shows the full-data logistic coefficients before and after, and the percentage change (after − before)/|before|.

Predictor	Before (full)	After (no priors)	% change	Shift in SEs
age	-0.044	-0.028	+34.9%	+5.58
sex_male	0.326	0.451	+38.4%	+1.74

juv_fel_count	0.072	0.441	+512.3%	+4.07
juv_misd_count	-0.012	0.387	+3,303.0%	+5.06
juv_other_count	0.274	0.380	+38.6%	+1.49
charge_felony	0.228	0.409	+79.0%	+3.07

The largest percentage shift is juv_misd_count (-0.012 → 0.387, +3,303%), although a near-zero start exaggerates it. In standard errors, age (+5.58 SEs, +34.9%), juv_misd_count (+5.06) and juv_fel_count (+4.07, +512%) all moved by more than four standard errors, and sex_male (+38.4%), juv_other_count (+38.6%) and charge_felony (+79.0%) also changed a lot. The same omission pushes the African-American race coefficient from 0.063 ($p = 0.318$) to 0.313 ($p < 0.001$), a +394% change, and cuts test accuracy from 67.9% to 62.4%.

What this shows. Priors correlate positively with the juvenile and charge variables, so the reduced model gives them credit for signal that belongs to priors. Juvenile misdemeanours went from irrelevant ($p = 0.878$) to "highly significant" ($p < 0.001$) just by leaving a variable out. A single fitted model gives no warning of this; every coefficient looks equally precise either way. So a model that was never stress-tested can show confident, significant-looking effects that are only artefacts of what was measured. I would trust a coefficient only after checking how stable it is when other variables are removed, and I would treat any variable that moves by more than one standard error as unreliable for policy conclusions.

Question 4. Detecting a hidden confound you cannot see

My proposal is an audit for candidate confounders, using external data and external validation, done before deployment and repeated every year. It comes from my result in 2.2: dropping one observed variable moved three coefficients by more than four standard errors and flipped two p-values from non-significant to significant, so a variable I never recorded could do the same without any sign.

Step 1 – Ask experts. Bring together public defenders, probation officers, a former judge and community representatives for a half-day workshop and ask one question: "What drives both who gets arrested and who re-offends, that this dataset does not record?" Likely answers are neighbourhood policing intensity, housing and job stability, and access to substance-use treatment. Rank them and write the list down before looking at any results.

Step 2 – Link outside data. Get the residence tract or zip from the court record and join public data: American Community Survey employment and poverty rates, and arrest density per person from the sheriff's office. These stand in for the suspected confounders.

Step 3 – Measure with the same yardstick. Refit Model B with each proxy added and flag any coefficient that moves by more than 1 SE (in 2.2 the shifts were between 1.49 and 5.58 SEs). Also regress the deviance residuals on each proxy: a significant slope means the model is missing information. Keep watching the calibration gap I found (zero-priors group: predicted 0.342 vs observed 0.288).

Step 4 – Validate elsewhere. Re-fit on a different county and a later cohort. If coefficients or calibration shift a lot, the model is picking up local confounds. Limits: this only finds confounds that someone thought of and could measure, it needs data-sharing agreements, and it takes weeks. That is why monitoring outcomes by subgroup after deployment must continue.

Question 5. The 50-row adversarial subset

Model B made 83 high-confidence mistakes (4.48% of the test set): 28 false positives and 55 false negatives. My random 50-row sample (seed 172) has 21 false positives and 29 false negatives. What they share is age: 92% of the 50 rows (46) are aged 30 or older versus 56.4% of the test set; 52% are over 45 versus 18.9%, and the mean age is 46.9 versus 34.5. Sex (80.0% male vs 80.5%) barely differs. Across all 83 confident errors, 76 (91.6%) are aged 30+, and the confident-error rate is 21.4% for ages 30+ versus 13.5% below 30.

The failure comes from the age coefficient (-0.0439 per year, OR 0.957), which adds up in a straight line: a 55-year-old is placed $30 \times 0.0439 = 1.32$ log-odds below a 25-year-old (odds × 0.27). With few priors (coefficient 0.162 per prior) this pushes probabilities below 0.25. The 29 false negatives are old (mean 55.0) and mostly have no record (55% zero priors). The 21 false positives have very long records (mean 14.8 priors) that push P above 0.75, which overstates risk because the effect of priors is curved (log(1+priors) LR = 64.1, $p = 1e\text{-}15$) and weaker for older people (age × priors coefficient -0.0022, $p = 0.0022$).

In statistical terms these are tail cases of an additive model that is linear in the log-odds. Part of the failure is unavoidable noise: the $P \leq 0.25$ group has a mean prediction of 0.184 yet 21.3% re-offend. The rest is over-confidence at the top (0.852 predicted vs 81.2% observed for $P \geq 0.75$), where the model has no interaction or curve to hold back extreme sums.

Question 6. What the trade-off cost means for a real person

Take two real test-set defendants, both men aged 43 who did not re-offend. Defendant A is Caucasian with 1 prior and a felony charge, model probability 0.315. Defendant B is African-American with 8 priors and a misdemeanour charge, probability 0.542. At $t = 0.35$ and $t = 0.45$, B is flagged high-risk and A is not; at $t = 0.55$ and above, neither is. B's record is longer, so the model is not "seeing race", but it does lean on priors, and the group-level data show who pays: at $t = 0.45$ the false-positive rate is 43.4% for African-American versus 28.2% for Caucasian non-re-offenders (ratio 0.651, which sets EO = 0.651). B is one of the 43% wrongly flagged, and he may face higher bail or a harsher sentence for something he would not have done.

Now put fairness through error rates first and raise the threshold to 0.65. The false-positive rate for B's group falls from 43.4% to 10.0% and B is no longer flagged. But two things go wrong. First, the ratio gets worse, not better: EO falls from 0.651 to 0.293 because the Caucasian rate collapses faster (28.2% to 2.9%).
Second, the true-positive rate for African-American defendants falls from 77.7% to 35.4%, so 65% of African-American defendants who will re-offend now go unflagged (up from 22%). They get no supervision or support, and their communities and eventual victims bear the cost.

The opposite choice hurts too. Putting demographic parity first by lowering the threshold to 0.25 brings DP to 0.838, but 89.6% of African-American non-re-offenders are flagged, and both defendants A and B are flagged (accuracy 51.7%). Equal-looking flag rates are reached by mislabelling almost everyone. No threshold removes the harm; it only moves it between people.

Question 7. A defensible threshold: t = 0.45

I choose t = 0.45, for three reasons. First, the fairness profile is balanced: DP = 0.615, EO = 0.651, PP = 0.819, so its worst ratio (0.615) is better than at every threshold from 0.55 to 0.95 (worst ratios 0.464 at 0.55 and 0.293 at 0.65). Second, it loses little accuracy: 66.6%, just 1.1 points below the best value (67.7% at t = 0.55), whereas t = 0.35, with a slightly better worst ratio (0.730), drops accuracy to 60.4% and flags 81% of African-American defendants. Third, the group rates are easy to read: it flags 37.2% of Caucasian and 60.5% of African-American defendants, and among the flagged people 52.5% (A) and 64.0% (B) actually re-offend, so a flag is right more often than not for both groups.

It is also defensible because the observed re-offending rates differ (37.3% vs 49.8% in the test set); with such base-rate differences, a threshold that flagged both groups equally would need a lot of error somewhere. At t = 0.45 the flag-rate gap (23.3 points) is about double the base-rate gap (12.5 points), which I accept only as a compromise.

Strongest counter-argument. A critic would say that at this threshold the false-positive rate is 43.4% for African-American but 28.2% for Caucasian non-re-offenders: 1.54 times the burden of wrongful flags on the group that is already over-represented in the data. That breaks equalized odds, the metric that tracks harm to innocent people, and EO = 0.651 is far from 1. I accept this is valid: t = 0.45 is a compromise, not a fair outcome, and it would need human review of flagged cases, a written rationale and regular re-auditing.

Question 8. Demographic parity or equalized odds?

I recommend equalized odds as the main criterion for recidivism risk scoring. The harm of a false positive (extra detention, a harsher sentence) falls on a specific person who would not have re-offended, and EO is the metric that compares that burden across groups. DP only compares raw flag rates and ignores what people actually did. With observed re-offending rates of 37.3% vs 49.8%, perfect DP would force the model to flag the groups equally, which means either under-flagging real re-offenders in one group or over-flagging non-re-offenders in the other.

My numbers show how hard this is in practice. In this data DP and EO rise and fall together, so one global threshold cannot deliver EO without destroying usefulness: EO is 0.804 at t = 0.25 but accuracy is 51.7%, below the 56.3% from flagging nobody. At t = 0.45, EO is 0.651 and accuracy 66.6%. Raising t from 0.45 to 0.65 improves PP by 19.8% but cuts EO by 55.0% and DP by 44.1%; lowering it to 0.25 lifts DP by 36.3% and EO by 23.6%, at an accuracy cost of 22.4%. So I would use EO as the monitored target, ask the policymaker to set a minimum acceptable EO, and use the model only as advice.

What I give up. Predictive parity, and to some extent demographic parity. At t = 0.45, PP is 0.819: 52.5% of flagged Caucasian defendants re-offend versus 64.0% of flagged African-American defendants. I accept this because, when base rates differ, EO and PP cannot both hold (Chouldechova 2017; Kleinberg et al. 2016), and a PP gap this size is what a well-calibrated model produces when the base rates are 37.3% vs 49.8%. DP = 0.615 also stays low. This is a value judgement, and policymakers should be able to see it and challenge it.

Part 4 – Reflection: "Just give me one number: deploy or not?"

1. Direct answer

Conditional deployment only: as advice for decision makers with human review, a stated threshold and yearly subgroup audits, never as an automatic decision-maker. The main reason is that the 67.9% test accuracy hides a false-positive rate of 43.4% for African-American versus 28.2% for Caucasian non-re-offenders (t = 0.45).

2. Why a single number fails

A policy-maker's "one number" (most likely overall accuracy or one fairness ratio) squeezes together several separate facts from my analysis, and each one changes the decision.

Accuracy hides subgroup gaps. The single figure of 67.9% test accuracy describes everyone at once, yet at t = 0.45 the model flags 60.5% of African-American defendants but only 37.2% of Caucasian ones, and wrongly flags 43.4% versus 28.2% of those who would not re-offend. An average cannot show who takes the errors.

The threshold is a values question, not a fact in the data. Accuracy peaks at t = 0.55 (67.7%), the best balance of the three fairness ratios among useful thresholds is around t = 0.35–0.45, and parity is "achieved" only at t ≤ 0.15 by flagging nearly everyone. Choosing between them means deciding how many wrongful flags are worth one prevented re-offence, and the data cannot decide that.

Fairness metrics conflict. Raising t from 0.45 to 0.65 improves predictive parity by 19.8% while cutting equalized odds by 55.0% and demographic parity by 44.1%. Because observed re-offending rates differ (37.3% vs 49.8%), no threshold satisfies all three, so any single fairness number quietly picks a side.

The model fails confidently on predictable groups. 83 test cases (4.5%) were confident mistakes, and 92% of my 50-row sample were aged 30 or over, a pattern traced to the linear age term and the missing age × priors interaction. An overall score cannot show that.

Hidden confounds stay unknown. Dropping a single variable (priors_count) moved the age coefficient by +5.6 standard errors and the African-American coefficient by +394%. If one observed variable can do that, an unmeasured one can do it with no visible warning, so no single number can prove the model is right for the right reasons. Only repeated stress-testing, as in Part 2, gives any evidence.

3. Evidence from my results

Part 1: 14.15% of people change class from the cut-off rule alone. Part 2.1: the juv_other_count coefficient is 0.553 in group A but 0.053 in group B. Part 2.2: juv_misd_count moved from -0.012 to 0.387. Part 2.3: 4.48% of test cases are confident errors. Part 2.4: at t = 0.45, DP = 0.615, EO = 0.651, PP = 0.819 and accuracy =
66.6%.

4. Closing

Responsible AI deployment needs openness about trade-offs, not a single metric. Show decision-makers the threshold, the subgroup error rates, the conflicting fairness metrics and the known failure regions; then let accountable people choose, explain their choice and keep monitoring.